

Tyler Weaver (he/him)

✉ maybe@tylerjw.dev ☎ (303) 903-5681 🌐 tylerjw 📄 tyler-weaver-b504626

Summary

Back-end Rust Engineer with production experience building a cloud-based archive and playback application in Rust, open-source robotics consulting, and wireless signal processing.

work experience

SciTec, Principal Software Engineer — IC on Archive and Playback Project in Rust Apr 2024 – present

- Async Axum HTTP web server to interface with Yugabyte database and file storage
- ZMQ playback node to stream data for a single host within an event
- File importing of binary data as C structs and Protobuf messages
- Reporting metrics to Prometheus using pull and push patterns
- Unit and integration testing
- Logging using tracing with Perfetto outputs for profiling
- GitLab CI pipelines for automating releases

PickNik Robotics, Staff Software Engineer — Assumed diverse responsibilities at fast-moving startup Apr 2019 – Feb 2024

- Authored Blog Series on Rust/C++ Interop: [Just the Basics](#), [CMake](#), [CXX](#)
- Maintainer of [Motion Planning Framework MoveIt](#) in the Robot Operating System (ROS) Ecosystem
- Author of Open-Source code-gen tool for ROS 2 parameters: [generate_parameter_library](#)
- Author of Library of C++17 Utilities: [ROS Support Library](#)
- Author and maintainer of documentation sites
- Author of feasibility studies for robotics startups
- Prepared and Led training on C++, Rust, Infrastructure, Testing, Debugging, SOLID, Functional Programming, Reading Code
- Technical lead for warehouse pick and place, onsite integration, hardware drivers, CI/CD, software/hardware testing
- Technical lead for fruit harvesting robot, onsite integration, localization, hardware drivers, CI/CD, optimization algorithm for registering vision results
- Developed Robotic truck refueling demo
- Developed Robotic satellite grapppling demo for Space Force grant
- Developed Firmware for two surgical robotic controllers, onsite integration, Conan/Docker/QNX
- CI/CD systems for OSS and internal projects using GitHub Actions, Gitlab, CircleCI
- CI systems using multiple compilers and static analysis tools
- CD systems using packaging (Conda, Conan, Debian) and containers
- [Maintainer and contributor to many OSS libraries in ROS](#)

Indiegogo funded project, Firmware Engineer May 2020 – Aug 2020

- [Thermal sensor for motorsports](#)
- Developed firmware for micro-controller application
- Math code written as a library that was also compiled and tested on the host
- CAN messaging to data-logger
- Optimized for code-space to fit on low-cost target

LGS Innovations, Software Engineer III

Apr 2014 – Mar 2019

- TS/SCI with polygraph
- Optimized C++ polyphase resampler and fir filter to run on ARM with NEON instructions
- Interface library for new SDR using Protobuf messaging and VRT over Ethernet
- Adversarial Research on 802.11b Physical Layer using an SDR
- Library for detecting oddly configured cellular basestations
- P25 scanner library
- Goertzel algorithm tuned for quick reject based on energy in frequency window, FSW timing and frequency recovery, Viterbi, RS, BCH, and CRC Decoding
- SDR interface libraries
- USRP N210, Sideiq
- MPT1327 scanner library
- IPP optimized fm demod, feedback timing recovery for MSK
- Ported Matlab library for unknown signal analysis into C++

TecStar Consulting, Software Engineer

Mar 2014 – June 2014

- Automation Testing using Selenium
- Visualization of Hadoop cluster jobs in JavaScript
- Google Glass Application

Colorado Army National Guard, Satellite Communication Systems Operator-Maintainer

Oct 2008 – Nov 2014

- Afghanistan deployment 2013

Education

BS DeVry University, Computer Engineering Technologies

2014

- Stereo vision system for Colorado Robotics Challenge
- Worked as a tutor for other Engineering students
- GPA: 3.96/4.0

Skills

Programming Languages: Rust, C++20, C++17, C++14, C, Python, Bash**Build Tools:** CMake, Makefiles, Cargo, Colcon, Bazel**CI/CD:** Github Actions, Gitlab-CI, Jenkins, SonarCloud, Docker, pre-commit**Packaging:** Pixi, Conda, Conan, Debian**Debugging:** GDB, Valgrind**Testing:** Selenium, Catch2, Gtest, Clang-Tidy, Clippy, ASan, TSan, MemSan**Libraries:** Eigen, Boost, GSL, Expected, Span, Protobuf, ROS, MoveIt**Racing:** Colorado Hill Climbs, Rally America, American Rally Association, World Racing League, National Auto Sport Association, Colorado Karting Tour